Jon Arrizabalaga

Postdoctoral Researcher at Massachusetts Institute of Technology

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Education

PhD **Technical University of Munich**, Robotics & Control • Thesis, Defense 2020 – 2025

MSc **KTH Royal Institute of Technology**, Mechatronics Engineering • Thesis 2018 – 2020

BSc **University of Navarre**, Mechanical Engineering 2014 – 2018

Academic Experience

Massachusetts Institute of Technology, Postdoctoral Researcher 2025 – present

Carnegie Mellon University, Visiting PhD / Postdoctoral Researcher 2024 – 2026

Charles University, Visiting PhD 2023

Shanghai Jiao Tong University, Visiting Student 2017 – 2018

Industry Experience

Bosch Research, MSc. Thesis - Robotics Researcher • Paper, Video 2020

Porsche AG, Research Intern 2019 – 2019

Honors & Awards

Best Robocup Paper Award (finalist) 2024
5 nominations out of ~4000 submissions at the conference IEEE/RJS IROS • Paper, Talk

Best PhD Lecturer Award 2022
Recognizes the PhD lecturer who has received the highest evaluations from BSc and MSc students

Outstanding End-of-Degree Award (finalist) 2018
Honors BSc. students who have excelled in both their academic and extracurricular endeavors

Invited Talks

Northeast Systems and Control Workshop, Columbia University 2025

Talking Robotics, Virtual Seminar • Talk 2025

GRASP Laboratory, Vijay Kumar Lab, University of Pennsylvania 2024

German Aerospace Center, DLR, Institute of Flight Systems 2023

Robotics and Perception Group, University of Zurich 2022

Teaching

Engineering Mechanics I, Lecturer, Technical University of Munich 2021 – 2024
Level: BSc, *Students*: 200 ~ 450

Mechanics for Aerospace, Lecturer, Technical University of Munich (Asia) 2021 – 2023
Level: MSc, *Students*: 20 ~ 25

Introduction to ROS (Robot Operating System), Lecturer, Technical University of Munich 2021 – 2022
Level: MSc, *Students*: 50

Dynamics Motion and Control, Teaching Assistant, KTH Royal Institute of Technology 2019 – 2020
Level: MSc, *Students*: 50

Robust Mechatronics, Teaching Assistant, KTH Royal Institute of Technology 2019 – 2020
Level: MSc, *Students*: 50

Publications

DQ-NMPC: Dual-Quaternion NMPC for Quadrotor Flight • Paper, Video, Code, Website — 2025
L.F. Recalde, D. Agrawal, *J. Arrizabalaga*, G. Li, RA-L

Convex Maneuver Planning for Spacecraft Collision Avoidance • Paper, Code — 2025
F. Vega, *J. Arrizabalaga*, R. Watson, Z. Manchester, IEEE iSpaRo

The Trajectory Bundle Method: Unifying Sequential-Convex Programming and Sampling-Based Trajectory Optimization • Paper — 2025
K. Tracy, J. Zhang, *J. Arrizabalaga*, S.Schaal, Y. Tassa, T. Erez, Z. Manchester, Under Review

Conflict-Based Search as a Protocol: A Multi-Agent Motion Planning Protocol for Heterogeneous Agents, Solvers, and Independent Tasks • Paper, Website — 2025
R. Veerapaneni,, A. Tang, H. He, S. Zhao, V. Shah, Y. Cen, Z. Ji, G. Olin, *J. Arrizabalaga*, Y. Shaoul, M. Likhachev Likhachev, Under Review

A Universal Formulation for Path-Parametric Planning and Control • Paper, Code, Website, Talk — 2025
J. Arrizabalaga, Z. Sir, Z. Manchester, M. Ryll, Under Review

PHODCOS: Pythagorean Hodograph-based Differentiable Coordinate System • Paper, Code, Talk — 2025
J. Arrizabalaga, F. Vega, Z. Sir, Z. Manchester, M. Ryll, IEEE Aero

Differentiable Collision-Free Parametric Corridors • *Best Paper Finalist* • Paper, Code, Video, Talk — 2024
J. Arrizabalaga, Z. Manchester, M. Ryll, IEEE IROS

Geometric Slosh-Free Tracking for Robotic Manipulators • Paper, Code, Video, Talk — 2024
J. Arrizabalaga, L. Pries, R. Laha, R. Li, S. Haddadin, M. Ryll, IEEE ICRA

Learning for CasADi: Data-Driven Models in Numerical Optimization • Paper, Code, Talk — 2023
T. Salzmann, *J. Arrizabalaga*, J. Andersson, M. Pavone, M. Ryll, L4DC

Pose-Following with Dual Quaternions • Paper, Code, Talk — 2023
J. Arrizabalaga, M. Ryll, IEEE CDC

SCTOMP: Spatially Constrained Time-Optimal Motion Planning • Paper, Video — 2023
J. Arrizabalaga, M. Ryll, IEEE IROS

Spatial Motion Planning with Pythagorean Hodograph curves • Paper, Video, Talk — 2022
J. Arrizabalaga, M. Ryll, IEEE CDC

Towards Time-Optimal Tunnel-Following for Quadrotors • Paper, Video — 2022
J. Arrizabalaga, M. Ryll, IEEE ICRA

Real-Time Neural-MPC: Deep Learning Model Predictive Control for Quadrotors and Agile Robotic Platforms • Paper, Code — 2022
T. Salzmann, E. Kaufmann, *J. Arrizabalaga*, M. Pavone, D. Scaramuzza, M. Ryll, RA-L

A caster-wheel-aware MPC-based motion planner for mobile robotics • Paper, Thesis, Video — 2021
J. Arrizabalaga, M. Ryll, IEEE ICAR

Students Mentored

Viraj Shah, Research Project, *now at* Carnegie Mellon University — 2025

Eden Cen, Research Project, *now at* Carnegie Mellon University — 2025

Akshay Jaitly, Research Project, *now at* Worcester Polytechnic Institute — 2025

Luis Fernando Recalde, Research Project, *now at* Worcester Polytechnic Institute — 2024

Giuseppe Soldati, Semester Thesis, *now at* German Aerospace Center (DLR) — 2023

Harun Tongay Tamtürk, Semester Thesis & MSc. Thesis, *now at* Airbus Defence and Space — 2023

Bruno Sorban, Research Project, *now at* Boston Consulting Group — 2023

Tommaso Faraci, Research Project, *now at* Universita di Trento — 2022

Extracurricular Projects

Ascent - Rocket Hopper, Guidance and Control Lead • Website 2022 – 2024

The Juggling Robot Project, Team Lead • Video 2019 – 2020

Formula Student Electric, Vehicle Dynamics Engineer 2016 – 2018

Review Activities

- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Systems, Man, and Cybernetics: Systems (T-SMC)
- Robotics Science and Systems (RSS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Conference on Decision and Control (CDC)
- IEEE International Conference on Humanoid Robots (Humanoids)
- IEEE International Conference on Space Robotics (iSpaRo)